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Neighbourhood blue space, health and wellbeing: The mediating role of different types of physical activity

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ABSTRACT

Background: Evidence suggests that living near blue spaces such as the coast, lakes and rivers may be good for health and wellbeing. Although greater levels of physical activity (PA) may be a potential mechanism, we know little about the types of PA that might account for this.

Objectives: To explore the mediating role of: a) 'watersports' (e.g. sailing/canoeing); b) 'on-land outdoor PA' in natural/mixed settings (e.g. walking/running/cycling); and, c) 'indoor/other PA' (e.g. gym/squash) in the relationships between residential blue space availability and health outcomes.

Methods: Using data from the Health Survey for England (n = 21,097), we constructed a path model to explore whether weekly volumes of each PA type mediate any of the relationships between residential blue space availability (coastal proximity and presence of freshwater) and self-reported general and mental health, controlling for green space density and a range of socio-economic factors at the individual- and area-level.

Results: Supporting predictions, living nearer the coast was associated with better self-reported general and mental health and this was partially mediated by on-land outdoor PA (primarily walking). Watersports were more common among those living within 5kms of the coast, but did not mediate associations between coastal proximity and health. Presence of freshwater in the neighbourhood was associated with better mental health, but this effect was not mediated by PA.

Conclusions: Although nearby blue spaces offer potentially easier access to watersports, relatively few individuals in England engage in them and thus they do not account for positive population health associations. Rather, the benefits to health from coastal living seem, at least in part, due to participation in land-based outdoor activities (especially walking). Further research is needed to explore the mechanisms behind the relationship between freshwater presence and mental health.

1. Introduction

A growing body of evidence suggests that, all else being equal, living near and regularly visiting the coast or other large water bodies (i.e. blue spaces), is associated with better general (Bauman et al., 1999; Burkart et al., 2016; Garrett et al., 2016; Garrett et al., 2019; Humpel et al., 2004; Wheeler et al., 2012) and mental (De Bell et al., 2017; Dempsey et al., 2018; Dzhambov et al., 2018; Gascon et al., 2017; Nutsford et al., 2016; Völker and Kistemann, 2011; de Vries et al., 2016; White et al., 2013a) health. In line with research into the potential health and wellbeing benefits of living near green spaces (e.g. parks and forests), at least three mechanisms might explain these relationships (Dadvand et al., 2016; Grellier et al., 2017; Hartig et al., 2014; James et al., 2016; Markevych et al., 2017; Triguero-Mas et al., 2015). The first potential mechanism is (physical) *harm reduction* where blue spaces, like other natural settings, are associated with lower levels of environmental stressors such as air pollutants and noise (Markevych et al., 2017), and less annoyance due to these stressors (Leung et al., 2017). A second mechanism relates to psychological *restoration*, which implies that blue and other natural environments may share restorative features that can help to alleviate psycho-physiological stress (Ulrich, 1983) and replenish attentional capacities (Kaplan and Kaplan, 1989). A third mechanism proposes that coastal and other aquatic settings may promote personal *capacity building* by encouraging physical activity

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社区蓝色空间、健康与福祉：其中的中介作用不同类型的体力活动

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ABSTRACT

背景：现有证据表明，居住在海岸、湖泊及河流等“蓝色空间”附近可能对健康与福祉有益。尽管增加体力活动（PA）水平可能是其中一种潜在机制，但关于哪些类型的体力活动可解释这一现象，目前仍知之甚少。

研究目的：探讨以下三个因素的中介作用：a) “水上运动”（如帆船/皮划艇）；b) 在自然/混合环境中进行的“陆地户外体力活动”（如步行/跑步/骑行）；以及 c) “室内/其他体力活动”（如健身房/壁球）——这些因素均作用于住宅区蓝色空间可及性与健康结局之间的关系。

方法：基于英格兰健康调查（Health Survey for England，n = 21,097）的数据，我们构建了一个路径模型，旨在探究每种体力活动（PA）类型的每周活动量是否介导住宅区蓝色空间可用性（沿海邻近性及淡水水体存在）与自我报告的总体及心理健康之间的任一关联，同时控制个体及区域层面的绿地密度及一系列社会经济因素。

研究结果：与预测结果一致，居住在沿海地区与受试者自我报告的更佳总体及心理健康状况相关，且这一关联部分由陆地户外体力活动（主要为步行）介导。水上运动在距海岸线5公里范围内的居民中更为常见，但并未介导沿海居住与健康之间的关联。社区内存在淡水资源与更佳心理健康状况相关，但该效应并非由体力活动介导。

结论：尽管邻近的蓝色空间可能使人们更容易参与水上运动，但在英格兰，参与此类活动的人群相对较少，因此这些空间并未展现出与人群健康相关的积极关联。相反，沿海居住对健康的益处，至少部分源于参与陆地户外活动（尤其是步行）的行为。未来仍需开展进一步研究，以探索淡水环境存在与心理健康之间关联的作用机制。

1. 引言

越来越多的证据表明，在其他条件相同的情况下，居住在沿海或其他大型水体（即“蓝色空间”）附近并定期前往这些区域，与更佳的整体健康（Bauman 等，1999；Burkart 等，2016；Garrett 等，2016；Garrett 等，2019；Humpel等，2004；Wheeler 等，2012）及心理健康（De Bell 等，2017；Dempsey 等，2018；Dzhambov 等，2018；Gascon 等，2017；Nutsford 等，2016；Volker 与 Kistemann，2011；de Vries 等，2016；White等，2013a）相关。基于关于居住在绿地（如公园与森林）附近可能带来的健康与福祉效益的研究，至少存在三种机制可解释这些关联：Dadvand 等，2016；Grellier 等，2017；Hartig 等，2014；James 等，2016；Markhevych等，2017；Triguero-Mas 等人，2015）。第一种潜在机制是（物理）危害减少，即蓝色空间与其他自然环境一样，与较低水平的环境压力源（如空气污染物和噪音）相关（Markevych等人，2017），以及因这些压力源引起的不适感较少（Leung 等人，2017）。第二种机制与心理恢复有关，这意味着蓝色及其他自然环境可能具备促进心理生理压力缓解（Ulrich，1983）及补充注意力能力（Kaplan 与 Kaplan，1989）的恢复特性。第三种机制则认为，沿海及其他水域环境可通过鼓励体力活动（PA）与增强社会凝聚力，促进个人能力建设（Markevych 等人，2017）。

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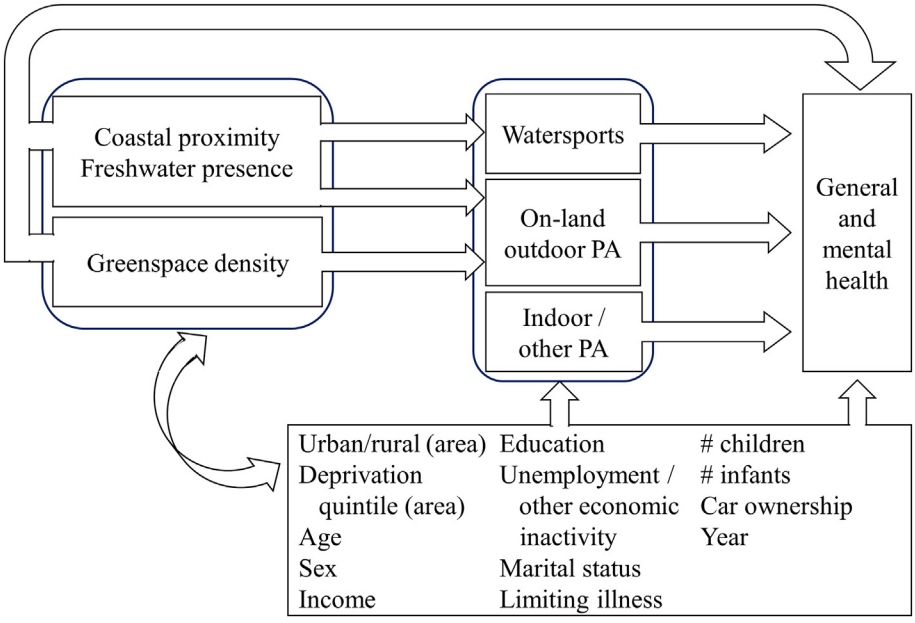


Fig. 1. Conceptual model in the present study.

(PA) and social cohesion (Markevych et al., 2017).

Although compelling, the capacity building mechanism, especially in terms of PA, lacks high quality epidemiological evidence to support it (Gascon et al., 2017; Markevych et al., 2017; van den Bosch and Ode Sang, 2017). A systematic review focusing on blue spaces in particular concluded that current evidence supports the idea that those who have better residential availability of blue space, most often measured by distance to the coast from the home, engage in more PA (Gascon et al., 2017). Regarding freshwater exposure, the evidence was too scant to draw conclusions, although inland waters such as rivers and lakes may afford recreational opportunities such as pathways for walking, similar to coastal settings (Gascon et al., 2017). Yet, we know little about the types of PA that are more commonly undertaken when living near different blue spaces (Gascon et al., 2017). Studies have rarely differentiated between activities occurring outdoors in, on, under, or by water and those conducted in indoor exercise facilities such as gyms in the local area (Gascon et al., 2017; Markevych et al., 2017). Therefore, it is uncertain if the explanatory role of physical activity in blue space-health relationships is limited to certain types of outdoor physical activities, or confounded by indoor physical activities.

However, tentative evidence from England suggests that visits to the coast may explain why people who live closer to the coast are more likely to engage in recommended levels of PA than those who live inland (White et al., 2014). Similarly, an Australian study found that residents in coastal locations reported on average 30 min more walking for exercise per week compared to those in non-coastal locations (Humpel et al., 2004). Furthermore, outdoor recreation patterns in England show that walking, by far the most common type of outdoor recreational activity in England, tends to last longer in coastal settings compared with urban green spaces (Elliott et al., 2015). Therefore, blue spaces may be relevant for public health by not only promoting PA in/on/under the water (i.e. watersports) but also land-based activities by water, such as coastal walking (Elliott et al., 2018), but this has not been extensively explored (White et al., 2016).

Insufficient levels of PA are a major health concern, both globally and within the UK (Hallal et al., 2012). Besides the well-established observation that most forms of PA are beneficial for both general and mental health (Fox, 1999), the setting of PA may provide additional effects. Experimental studies consistently suggest that outdoor PA brings about greater affective benefits than indoor PA (Thompson Coon et al., 2011). Similarly, observational evidence has shown that conducting regular nature-based PA is more strongly connected to some aspects of positive mental health compared to built outdoor or indoor settings (Mitchell, 2013; Pasanen et al., 2014). Further, coastal visits are associated with greater stress reduction (White et al., 2013b), happiness (MacKerron and Mourato, 2013), and connectedness to nature (Wyles et al., 2019) than other natural settings, so it seems plausible that physical activity in such a setting, including both watersports per se, as well as on-land activities such as walking near the water, may be particularly beneficial.

The current research explored these issues using data from a representative sample of the population of England. We examined the relationships between coastal proximity (how close someone lives to the sea) and freshwater presence with self-reported health and well-being, as a function of different types of PA in outdoor and indoor settings: a) in/on/under the water (i.e. *watersports*); b) *on-land outdoor PA* conducted in natural/mixed settings, possibly also by the water (e.g. walking, running and cycling); and c) *indoor/other PA*, conducted in indoor/built settings specifically designed for PA (gyms, swimming pools, sports centres etc.). Considering the importance of walking for public health (Hamer and Chida, 2008; Lee and Buchner, 2008; Robertson et al., 2012), additional analyses separated walking from other on-land outdoor activities. Indoor PA was not assumed to be related to blue or green space due to a lack of theoretical justification for these connections, although it is possible that in greener, coastal, or more rural settings there may be less indoor facilities for PA. Using the theoretical structure shown in Fig. 1, we constructed a path model to explore the mediating role of these different types of outdoor PA blue space – well-being relationships, controlling for potential confounders including green space density. This approach directly addresses one of the research priorities indicated by recent systematic reviews on the topic of blue/green spaces and health (Gascon et al., 2017; Markevych et al., 2017).

2. Material and methods

2.1. Data

The Health Survey for England (HSE) is an annual cross-sectional survey, typically surveying at least 8000 individuals throughout the year (adults and children). The sampling procedure uses a multi-stage stratified probability sampling design for private households in England

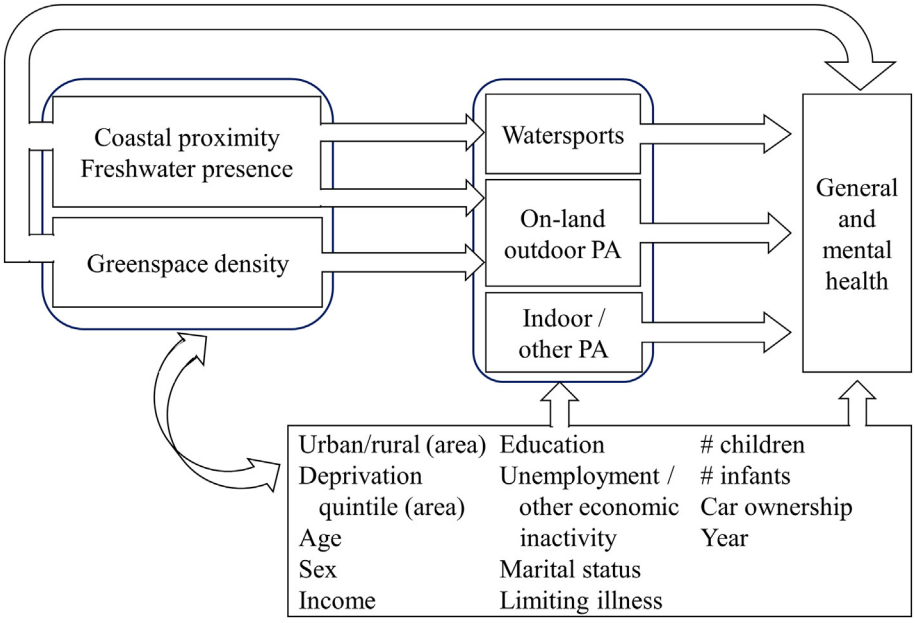


图 1. 本研究的概念模型。

尽管这一能力建设机制颇具说服力，但在体力活动（PA）方面，仍缺乏高质量流行病学证据予以支持（Gascon等人，2017；Markhevych等人，2017；van den Bosch与Ode Sang，2017）。一项专门针对蓝色空间开展的系统综述得出结论：现有证据支持以下观点——居住环境中蓝色空间资源更丰富的个体（通常以住宅距海岸线的距离衡量）更倾向于进行体力活动（Gascon等人，2017）。关于淡水接触，现有证据仍过于匮乏，尚无法得出明确结论，尽管河流与湖泊等内陆水域可能提供与沿海环境相似的休闲机会（如步行路径）（Gascon 等人，2017）。然而，对于居住在不同蓝色空间附近时，个体更常进行的体力活动类型，我们仍知之甚少（Gascon 等人，2017）。研究很少区分在户外、水上、水下或水中进行的活动与在当地室内健身场所（如健身房）中进行的活动（Gascon 等，2017；Markevych 等，2017）。因此，尚不确定体力活动对蓝色空间健康关系的解释作用是否仅限于某些类型的户外体力活动，还是受到室内体力活动的干扰。

然而，来自英格兰的初步证据表明，前往海岸地区可能解释了为何居住在靠近海岸地区的人比居住在内陆地区的人更有可能达到推荐的体力活动水平（White 等，2014）。同样，一项澳大利亚研究发现，与非沿海地区的居民相比，沿海地区的居民每周平均多步行30分钟用于锻炼（Humpel 等，2004）。此外，英格兰的户外休闲活动模式显示，步行——英格兰最常见的户外休闲活动类型——在沿海环境中比在城市绿地中往往持续时间更长（Elliott 等，2015）。因此，蓝色空间可能对公共健康具有重要意义，不仅能促进水中或水边活动（如水上运动），还能促进以水为媒介的陆地活动，例如沿海步行（Elliott 等，2018），但这一方面尚未得到广泛研究（White 等，2016）。

体力活动水平不足已成为全球及英国境内的一大健康隐患（Hallal 等，2012）。除了已充分证实的、大多数形式的体力活动对整体及心理健康均有益这一观点外（Fox，1999），体力活动的环境设置还可能带来额外效益。实验研究一致表明，户外体力活动比室内体力活动带来更多积极效益（Thompson Coon等，2011）。同样，观察性证据也显示，与人工构建的户外或室内环境相比，定期进行基于自然的体力活动与积极心理健康的某些方面关联更为密切（Mitchell，2013；Pasanen 等，2014）。此外，相较于其他自然环境，前往沿海地区与更大的压力缓解（White 等，2013b）、幸福感提升（MacKerron 与 Mourato，2013）以及与自然联结感增强（Wyles 等，2019）相关，因此，在此类环境中进行的体力活动——包括水上运动本身以及近水步行等陆地活动——可能具有特殊价值。早期即显现益处。

本研究利用英格兰人口代表性样本的数据探讨了这些问题。我们考察了沿海居住距离（个人居住地与海洋的距离）及淡水资源存在情况与自我报告的健康及福祉之间的关系，并考虑了户外与室内环境中不同类型体力活动的影响：a) 水中/水上/水下活动（即水上运动）；b) 陆地户外体力活动，在自然/混合环境中进行，也可能涉及水域活动（如步行、跑步和骑行）；以及 c) 室内或其他体力活动，在专为体力活动设计的室内/建筑环境中进行（如健身房、游泳池、体育中心等）。鉴于步行对公共健康的重要性（Hamer 与 Chida，2008；Lee 与 Buchner，2008；Robertson 等，2012），我们进一步将步行与其他陆地户外活动分开分析。对于室内体力活动，由于缺乏这些关联的理论依据，我们未假设其与蓝色或绿色空间有关，尽管在更绿意盎然、沿海或更偏农村的环境中，可供使用的室内体力活动场所可能较少。本研究采用如下理论框架：图1中，我们构建了一个路径模型，以探究不同类型户外体力活动蓝色空间与幸福感之间的中介作用，同时控制包括绿地密度在内的潜在混杂变量。这一方法直接回应了近期关于蓝色/绿色空间与健康主题的系统综述中所指出的研究重点之一（Gascon 等，2017；Markevych 等，2017）。

2. 材料与方法

2.1. 数据

英格兰健康调查（HSE）是一项年度横断面调查，通常全年对至少8000名个体（成人及儿童）进行抽样调查。其抽样程序采用多阶段分层概率抽样设计，针对英格兰的私人家庭（Bridges 等，2013）。在2008年

Table 1
Summary of the categorisation of different types of leisure time PA.

Intensity	Watersports (Papathanasopoulou et al., 2016)	On-land outdoor PA (Mytton et al., 2012)	Indoor/other PA
Light (2 METs/h)	– Fishing, surfing/wind surfing, sailing, angling	– Walking at an average or a slow pace	– Dancing and "exercises" (if were not out of breath/sweaty) – All 'other' light activities
Moderate (4.5 METs/h)	– Fly fishing, snorkelling, water skiing, canoeing, kayaking	– Walking at a brisk or fast pace – Cycling and football/rugby (if were not out of breath/sweaty)	– Dancing and "exercises" (if were out of breath/sweaty) – Swimming, working out/exercise bike/weight training, aerobics or similar, and badminton/tennis (if were not out of breath/sweaty) – All 'other' moderate activities
Vigorous (9 METs/h)	– Scuba diving, subaqua diving and rowing	– Running/jogging – Cycling and football/rugby (if were out of breath/sweaty)	– Squash – Swimming[a], working out/exercise bike/weight training, aerobics or similar, and badminton/tennis (if were out of breath/sweaty) – All 'other' vigorous activities

[a] Swimming was not distinguished as being indoors in a pool or outdoors in open water; given the likelihood that most swimming is in indoor pools, swimming was classified as indoor PA.

(Bridges et al., 2013). In both 2008 and 2012, selected households were initially approached with an advance letter containing information about the survey, after which trained interviewers made contact to arrange a computer-assisted, face-to-face interview with all adults within the same household (Bridges et al., 2013). More specific details of data collection, sampling, and descriptive statistics are well-documented elsewhere (Bridges et al., 2013). We pooled two waves of the data series, 2008 and 2012, that included additional questions on the amount and types of PA undertaken in the past four weeks. The sample in 2008 consisted of a 'booster' sample in addition to the core sample with twice as many respondents in total (Aresu et al., 2009b). In 2012, the sample consisted of 8291 individuals in 5219 households (Bridges et al., 2013), and in 2008, the sample consisted of 15,102 individuals in 9191 households (Aresu et al., 2009b). Due to missing values, mainly in the outcomes (Table 2), the main models analysed here included 21,097 respondents.

2.2. Measures

2.2.1. Area-level environmental indicators

All area-level variables were specified at the Lower-layer Super Output Area (LSOA) level. LSOAs are small geographical areas with an average of approximately 1500 residents in 600 households, and cover, on average, 4 km^2. This unit of analysis is in line with the recommendations to use larger areas than immediate surroundings when assessing the connection between environmental features and PA (Browning and Lee, 2017; Markevych et al., 2017). Since standard HSE datasets only include larger geographical region identifiers, the data owners permitted anonymised linkage of three environmental variables with the approval of the Data Release Panel and the Health and Social Care Information Centre (now NHS Digital). LSOA-level environmental data were provided to the data custodian, where they were linked based on the participant's LSOA of residence and returned without spatial identifiers. In order that residential location could not be reverse-engineered, continuous measures of green and blue space were classified into relatively coarse categorical variables before being supplied to the data custodian, and the returned linked data was stripped of broader regional identifiers.

Coastal proximity was calculated as the Euclidean distance from the population-weighted centroid of the respondent's LSOA to the nearest coastline as derived from open license geographical datasets (Office for National Statistics Geography, 2016). Distances were collapsed into five categories: 0–1 km, > 1–5 km, > 5–20 km, > 20–50 km, and > 50 km consistent with previous research (e.g. Wheeler et al., 2012). To obtain approximately equal sample sizes per category, we combined the first two categories (0–5 km) for current analyses.

Freshwater coverage is relatively low in England compared to other EU countries (overall land coverage 1.3%, Morton et al., 2011), and accordingly, we modelled it as either none or at least some (> 0%), indicating the absence or *presence of freshwater* in the LSOA. This variable was derived from the CEH Land Cover Map 2007 (Morton et al., 2011).

Green space area density (%) within an individual's neighbourhood LSOA was provided to us in 10% bands (< 10%, 10– < 20%, etc.), derived from the Generalised Land Use Database (GLUD) from openly available neighbourhood statistics (Department for Communities and Local Government, 2007). These data include all green spaces (irrespective of public/private accessibility) of minimum size 5 m^2, excluding domestic gardens, and have been used in previous relevant research (Mitchell and Popham, 2007; Wheeler et al., 2012). We re-scaled the categories to 1–5 where one scale point increase refers to 20% increase in green space density to ease interpretation.

2.2.2. Leisure-time PA: watersports, on-land outdoor PA, indoor/other PA

Respondents were asked to report all leisure-time PA conducted in the past four weeks. Regarding each activity, they recorded the number of days it was conducted, average duration per bout, and whether the activity had usually made them 'out of breath and sweaty'; this comprehensive instrument has been developed for accurate assessments of whether respondents meet current PA recommendations (Scholes and Mindell, 2013). Following the official documentation of the HSE (Bridges et al., 2013), each activity was classified in terms of its intensity as light, moderate or vigorous, depending on the type of activity and whether it made respondents short of breath (summarised in Table 1). To calculate proxies for the intensities for each activity, we ascribed Metabolic Equivalents of Task (METs; Ainsworth et al., 2011) rates so that the hours spent on all light activities were multiplied by 2, moderate activities by 4.5, and vigorous activities by 9. This approach reflects the public health recommendations where conducting vigorous activities are considered to be twice as beneficial for health as moderate activities (Scholes and Mindell, 2013).

Using these approaches, we calculated per-person estimates of total MET-h/week, based on previous studies that also used HSE data, for watersports (Papathanasopoulou et al., 2016), on-land outdoor PA (Mytton et al., 2012), and indoor/other physical activities (Table 1). A key sensitivity analysis divided on-land outdoor PA into 'walking' and all other on-land outdoor activities (namely running, cycling, football/rugby). In line with the HSE methods documentation, which offers best practice guidance on avoiding undue influence of outliers on any analyses (NatCen Social Research, 2014), the maximum weekly hours for each PA subcategory was truncated to 40. In these cases, the respective MET hours were recalculated by multiplying 40 by the respondents' average PA intensity in that subcategory.

2.2.3. General and mental health

General health was asked with a single item "How is your health in

表 1
各类休闲时间体力活动分类概述

强度	水上运动（Papathanasopoulou 等，2016）	陆地户外体力活动（Mytton 等人，2012）	室内/其他体力活动
光照强度（2 METs/h）	_钓鱼、冲浪/风帆冲浪、帆船运动，钓鱼	以中等或缓慢速度行走 速度	_跳舞及“锻炼”（若未气喘/出汗）所有 ζother 的光学活性
中等强度（4.5 METs/h）	飞钓、浮潜、滑水、划独木舟、划皮划艇	_以快走或快走速度行走 _骑行及足球/橄榄球运动（若未感到气喘或出汗）	_跳舞及“锻炼”（若感到气喘/出汗）游泳、健身/动感单车/力量训练、有氧运动或类似活动，以及羽毛球/网球（若未感到气喘或大汗淋漓）所有 ζother 的常规活动
高强度运动（9 METs/h）	水肺潜水、亚水潜水及划桨运动	_跑步/慢跑_ _骑行及足球/橄榄球运动（若感到呼吸急促或出汗）	_南瓜 游泳、健身/动感单车/力量训练、有氧运动或类似活动，以及羽毛球/网球（若感到呼吸急促或汗出）所有 ζother 的活跃活动

[a] 游泳活动并未被区分为在室内泳池中进行或在室外开放水域中进行；鉴于大多数游泳活动均在室内泳池中进行，因此将游泳归类为室内体力活动。

和2012年，首先向选定家庭发送一封预先信函，其中包含有关该调查的信息；随后由经过培训的访谈员联系这些家庭，安排与同一家庭中的所有成人进行计算机辅助的面对面访谈（Bridges 等，2013）。关于数据收集、抽样及描述性统计的更多具体细节，已在其他文献中详细记录（Bridges 等，2013）。我们将2008年与2012年的数据系列合并，这两次调查均增加了关于过去四周内进行的体力活动（PA）的数量与类型的附加问题。2008年的样本除核心样本外，还包含一个ζbooster样本，总受访者数量增至两倍（Aresu 等，2009b）。2012年，样本包含来自5219户家庭的8291名个体（Bridges等，2013）；2008年，样本包含来自9191户家庭的15,102名个体（Aresu 等，2009b）。由于存在缺失值（主要出现在结果变量中，见表2），本文分析的主要模型共包含21,097名受访者。

2.2. 措施

2.2.1. 区域级环境指标

所有区域级变量均在下层超级输出区（LSOA）层面进行设定。LSOA 是小型地理区域，平均包含约 1500 名居民和 600 户家庭，平均覆盖面积为 4 平方公里。这一分析单位符合相关建议，即在评估环境特征与身体活动（PA）之间的关联时，应采用比直接周边区域更大的区域（Browning 与 Lee，2017；Markevych 等，2017）。由于标准 HSE 数据集仅包含较大地理区域标识符，数据所有者在获得数据发布小组及健康与社会护理信息中心（现为 NHS Digital）的批准后，允许将三个环境变量进行匿名关联。 LSOA 层级的环境数据被提供给数据托管方，随后根据参与者的居住 LSOA 进行关联，并在返回时去除空间标识符。为防止住宅区位信息被逆向工程破解，研究人员先将绿地与蓝绿空间的连续测量值划分为相对粗粒度的分类变量，再将其提交给数据管理方；随后，从返回的关联数据中剔除更广泛的区域标识符。

沿海邻近性是通过计算受访者所在 LSOA 区域的人口加权质心与最近海岸线之间的欧氏距离来衡量的，该距离数据源自开放许可的地理数据集（英国国家统计局地理部门，2016）。距离被划分为五个区间：0–1公里、>1–5公里、>5–20公里、>20–50公里以及>50公里，此分类方式与既往研究一致（例如Wheeler等，2012）。为在每类区间中获得近似相等的样本量，在本次分析中我们将前两个区间（0–5公里）合并处理。

与其他欧盟国家相比，英格兰的淡水覆盖面积相对较低（总体陆地覆盖率为1.3%，Morton等人，2011），因此，我们将该变量建模为“无”或“至少部分存在”（>0%），以表示 LSOA 区域内是否存在淡水。该变量源自《 CEH 2007年土地覆盖图》（Morton等人，2011）。

个人居住区内的绿地面积密度（%）LSOA 以 10% 为区间（<10%、10–<20% 等）提供给我们，这些数据源自通用土地利用数据库（GLUD），该数据库基于公开可用的社区统计数据（社区与地方政府部，2007年）构建而成。这些数据包含所有绿地（不可从公共/私人可及性的角度，最小尺寸为5米，不包括家庭花园，且已在既往相关研究中应用（Mitchell 与 Popham，2007；Wheeler 等，2012）。我们将这些类别重新划分为1–5级，其中每个等级点的增加代表绿地密度增加20%，以便于解读。

2.2.2. 闲暇时间体力活动：水上运动、陆地户外体力活动、室内及其他体力活动

受访者被要求报告过去四周内进行的所有休闲时间体力活动。针对每项活动，他们记录了开展该活动的天数、每次活动的平均持续时间，以及该活动是否通常会导致其呼吸急促并出汗；此项信息已开发出一套综合性工具，用于准确评估受访者是否符合现行体力活动（PA）推荐标准（Scholes 与 Mindell，2013）。遵循 HSE 官方文档（Bridges 等，2013）的规定，每项活动均根据其强度被划分为轻度、中度或剧烈三类，具体分类依据活动类型以及是否导致受访者出现呼吸急促（详见表 1）。为计算每项活动的强度代用指标，我们采用了任务代谢当量（METs；Ainsworth 等，2011）换算率：所有轻度活动所花费的小时数乘以 2，中度活动乘以 4.5，剧烈活动乘以 9。此方法体现了公共卫生建议——进行剧烈活动对健康的益处是中等强度活动的两倍（Scholes 与 Mindell，2013）。

采用这些方法，我们基于先前同样使用 HSE 数据的研究，分别计算了水上运动（Papathanasopoulou 等，2016）、陆地户外体力活动（Mytton 等，2012）以及室内/其他体力活动（表1）中每人每周总MET时数的估计值。一项关键敏感性分析将陆地户外体力活动分为“ζ步行”与所有其他陆地户外活动（即跑步、骑行、足球/橄榄球）。参照 HSE 方法文档中关于避免异常值对任何分析产生不当影响的最佳实践指南（NatCen Social Research，2014），每个体力活动子类别的每周最大活动时长均被截断至40小时。在此情况下，相应的MET时数通过将40小时乘以受访者在该子类别中的平均体力活动强度重新计算得出。

2.2.3. 全身及心理健康

总体健康状况通过单一问题“您总体健康状况如何？”进行询问，

general?" and assessed on a 5-point ordinal scale ranging from 1 'Very bad' to 5 'Very good'. This is a widely used and validated measure for assessing perceived general health (Doiron et al., 2015; Mavaddat et al., 2011) that correlates with numerous objective health outcomes (e.g. mortality; Kyffin et al., 2004) and has been used extensively in previous green/blue space research (e.g. Mitchell and Popham, 2007; Wheeler et al., 2012).

Mental health was assessed with the General Health Questionnaire (GHQ-12; Goldberg et al., 1997). The scale contains 12 items related to issues with mental health in the past few weeks (for example, 'How often have you been feeling unhappy and depressed?'), measured on a 4-point scale ranging from 0 'not at all' to 4 'much more than usual'. GHQ has been widely used in health surveys and in other studies of blue space and mental health (e.g. White et al., 2013a) and thus, to enable comparisons across studies it has been recommended to be used in people-environment studies (Gascon et al., 2015). While we recognize that GHQ-12 scores are a latent construct dependent on the 12 items the questionnaire comprises, we a priori modelled the overall 0–12 score which is in line with established scoring conventions (Goldberg and Williams, 1988) and facilitates comparison with previous research. The scores were inverted so that higher values indicated better mental health. The internal consistency of the scale, measured by Cronbach's α, was high (0.898).

2.2.4. Covariates

We adjusted the analyses for two area-level covariates that may play a role on the quantity and quality of available blue and green space (Markevych et al., 2017; van den Berg et al., 2015): *urban/rural status* (rural including towns, fringes, villages, hamlets, or isolated dwellings) and *deprivation*, measured as the quintile of the Index of Multiple Deprivation (QIMD; Department for Communities and Local Government, 2008, 2011).

In terms of individual- and household-level covariates, we adjusted the analyses for a number of demographic and socio-economic indicators that may influence health or health behaviour based on recommendations from previous literature (van den Berg et al., 2015; Gascon et al., 2015; Gascon et al., 2017; Giles-Corti and Donovan, 2003; Lee and Buchner, 2008). *Age*, *sex*, and *education* (Table 2) could confound blue space – health relationships although supporting evidence is mixed (Bélanger et al., 2011; van den Berg et al., 2015; Gascon et al., 2015; Gascon et al., 2017). *Marital status* can reflect both mental health and PA patterns (e.g. by having a significant other to walk with; Lee and Buchner, 2008). Annual *household income*, being *unemployed* or otherwise *economically inactive* (such as retired or a stay-at-home parent), and *car availability* can influence the green and blue settings one has access to, how much time they have to spend on PA there, and the types of PA they can afford to engage in (van den Berg et al., 2015). The number of *children* or *infants* can have an influence on the quantity and outcomes of visits to green or blue environments (White et al., 2013b). Having a *long-term limiting illness* is directly linked to health and mental health and the ability to visit outdoor spaces (Boyd et al., 2018). Of note, although this variable is often included in such models it may act as a potential mediator in its own right and, hence, confound the associations between environmental indicators, PA, and health. Preliminary analyses, however, found no evidence for confounding effects so it was retained only as a covariate. Finally, as the survey was conducted in two separate waves, we adjusted the analyses for *year* (2008 or 2012) to account for any potential national differences in health behaviour or outcomes.

2.3. Analytical approach

In line with recommendations to use path modelling tools to assess the environment-health mechanisms (Markevych et al., 2017) and recent related work on the topic (Dzhambov et al., 2018), we analysed all our relationships of interest in a single path model using the 'lavaan

Table 2
Sample descriptives. All cases with missing information were excluded from the main analysis (n = 2291, 9.8% of the sample).

Variable	(Category)	*n*/frequency	Mean/ % of total	*SD*
General health (1–5)		23,379[b]	3.97	0.95
Mental health (0–12)		21,474[c]	10.6	2.64
Physical activity (MET h/ week)	Watersports	23,388	0.10	1.76
	On-land outdoor PA	23,388	14.58	25.61
	Indoor/other PA	23,388	7.21	17.28
Coastal proximity (km)	> 50[a]	8840	37.8	
	> 20–50	6611	28.27	
	> 5–20	3927	16.79	
	0–5	4010	17.15	
Freshwater presence	None[a]	17,956	76.77	
	Some (> 0%)	5432	23.23	
Green space density (%)	0– < 20	6345	27.13	
	20– < 40	5742	24.55	
	40– < 60	3563	15.23	
	60– < 80	3086	13.20	
	≥80	4652	19.89	
Urban/rural status (area)	Urban[a]	18,447	78.87	
	Town & fringe, village, hamlet or isolated dwelling	4941	21.13	
Deprivation (QIMD quintile, area)	1st - Least deprived[a]	5141	21.98	
	2nd	4792	20.49	
	3rd	4676	19.99	
	4th	4517	19.31	
	5th - Most deprived	4262	18.22	
Age (years)	16–24	2538	10.85	
	25–34	3355	14.34	
	35–44	4097	17.52	
	45–54	3908	16.71	
	55–64	3862	16.51	
	65–74	3109	13.29	
	≥75	2519	10.77	
Sex	Female[a]	12,949	55.37	
	Male	10,439	44.63	
Equivalised annual household income (in thousands), excluding missing values		18,674	32.42	29.23
Income missing	No[a]	18,674	79.8	
	Yes	4714	20.2	
Education (highest level)	No qualification[a]	5919	25.31	
	NVQ1/CSE, other grade equivalent	1087	4.65	
	NVQ2/GCE, O Level equivalent	4906	20.98	
	NVQ3/GCE, A Level equivalent	3462	14.8	
	Higher education below degree	2577	11.02	
	NVQ4/NVQ5/ Degree or equivalent	4972	21.26	
	NA/no answer/ refused	465	1.99	
Economic activity	In employment/ student[a]	13,097	97.32	
	Unemployed	627	2.68	
	Retired or other economically inactive	9664	41.32	
Long-standing limiting-illness	No[a]	17,567	75.11	
	Yes	5821	24.89	
Marital status	Other[a]	8500	36.34	
	Married, in cohabitation, or civil partnership	14,888	63.66	
Children (aged 2–15)	None	17,480	74.74	
	1	3013	12.88	
	≥2	2895	12.37	

(*continued on next page*)

并采用从1分（ζ极差）到5分（ζ极佳）的5级有序量表进行评估。该量表是评估个体感知总体健康状况时广泛使用且经过验证的工具（Doiron等人，2015；Mavaddat等人，2011），其与众多客观健康结局指标相关（如死亡率；Kyffin等人，2004），并在既往绿色/蓝色空间研究中被广泛应用（例如Mitchell与Popham，2007；Wheeler等人，2012）。

心理健康状况采用一般健康问卷（GHQ-12；Goldberg等人，1997）进行评估。该量表包含12个与过去几周内心理健康相关的问题（例如，“您感到不快乐和抑郁的频率是多少？”），采用4点量表进行测量，评分范围从0（“完全没有”）到4（“比平时多”）。GHQ已被广泛应用于健康调查及其他关于蓝色空间与心理健康的研究中（如White等人，2013a），因此，为了便于不同研究之间的比较，建议在人群-环境研究中使用该量表（Gascon 等人，2015）。尽管我们承认GHQ-12的得分是一个依赖于该问卷所包含的12个问题的潜在构念，但我们仍先验地对总体0–12分进行了建模，这与既有的评分规范一致（Goldberg 与 Williams，1988），并有助于与既往研究进行比较。得分经过反转处理，使得较高数值代表更好的心理健康状况。该量表的内部一致性（以Cronbach ‘s α系数衡量）较高（0.898）。

2.2.4. 协变量

我们对分析进行了两项区域层面协变量的调整，这些协变量可能对可用蓝色与绿色空间的数量及质量产生影响（Markevych 等，2017；van den Berg 等，2015）：城市/乡村属性（乡村区域包括城镇、边缘地带、村庄、小村落或孤立住宅）以及贫困程度——后者以多重贫困指数（QIMD；社区与地方政府部，2008,2011）的五分位值衡量。

在个体及家庭层面的协变量方面，我们根据既往文献建议（van den Berg等，2015；Gascon等，2015；Gascon等，2017；Giles-Corti与Donovan，2003；Lee与Buchner，2008），对分析结果进行了若干可能影响健康或健康行为的人口统计学及社会经济指标校正。年龄、性别及教育程度（表2）可能干扰“蓝色空间—健康”关联，但支持性证据尚不一致（Belanger等，2011；van den Berg等，2015；Gascon等，2015；Gascon等，2017）。婚姻状况既可反映心理健康状况，也可反映体力活动模式（例如拥有可共同散步的伴侣；Lee与Buchner，2008）。家庭年收入、是否失业或处于其他经济非活跃状态（如退休或全职育儿者），以及是否拥有汽车，都可能影响个体可接触的绿色或蓝色环境、在这些环境中进行体力活动的可用时间，以及可参与的具体体力活动类型（van den Berg et al.，2015）。子女或婴儿的数量也会影响访问绿色或蓝色环境的次数及结果（White et al.，2013b）。患有长期限制性疾病与健康及心理健康状况，以及访问户外空间的能力直接相关（Boyd et al.，2018）。值得注意的是，尽管这一变量常被纳入此类模型，但它本身也可能作为潜在中介变量发挥作用，从而混淆环境指标、体力活动与健康之间的关联。然而，初步分析未发现存在混杂效应的证据，因此仅将其作为协变量保留。最后，由于调查分两波进行，我们根据年份（2008年或2012年）对分析结果进行了调整，以考虑全国范围内健康行为或结果可能存在的差异。

2.3. 分析方法

根据使用路径建模工具评估环境与健康机制的建议（Markevych 等，2017）以及该主题领域的最新相关研究（Dzhambov 等，2018），我们使用 ζlavaan 在一个单一路径模型中分析了所有感兴趣的变量关系。

表 2
样本描述性统计。所有信息缺失的病例均被排除在主要分析之外（n = 2291，占样本量的 9.8%）。

变量	（类别）	n/频率	平均值/占总值的百分比	*SD*
总体健康状况 (1–5) 心理健康状况 (0–12) 体力活动水平 (MET·h)		23,379[b]	3.97	0.95
		21,474[c]	10.6	2.64
一星期)	水上运动	23,388	0.10	1.76
	陆地户外体力活动 / 室内或其他体力活动	23,388	14.58	25.61
		23,388	7.21	17.28
距海岸线距离（公里）	＞ 50[a]	8840	37.8	
	＞ 20–50	6611	28.27	
	＞ 5–20	3927	16.79	
	0–5	4010	17.15	
淡水分布	无[a]	17,956	76.77	
	部分（＞0%）	5432	23.23	
绿地密度（%）	0– ＜ 20	6345	27.13	
	20– ＜ 40	5742	24.55	
	40– ＜ 60	3563	15.23	
	60– ＜ 80	3086	13.20	
	≥80	4652	19.89	
城乡属性（区域）	城市[a]	18,447	78.87	
	城镇及边缘地带、村庄、小村落或独立住宅	4941	21.13	
剥夺程度（QIMD 五分位区间，区域）	第一 - 最小被剥夺[一项]	5141	21.98	
	第2	4792	20.49	
	第3	4676	19.99	
	第4	4517	19.31	
	第 5 位 - 最贫困群体	4262	18.22	
年龄（岁）	16–24	2538	10.85	
	25–34	3355	14.34	
	35–44	4097	17.52	
	45–54	3908	16.71	
	55–64	3862	16.51	
	65–74	3109	13.29	
	≥75	2519	10.77	
性别	女性[a]	12,949	55.37	
	男性	10,439	44.63	
标准化年度家庭收入（单位：千元），不含缺失值		18,674	32.42	29.23
收入缺失	否[a]	18,674	79.8	
	是	4714	20.2	
教育（最高学历）	无资格[a]	5919	25.31	
	NVQ1/CSE，其他	1087	4.65	
	相当于 NVQ2/GCE 的等级、相当于 NVQ3/GCE 的 O Level 等级、相当于 A Level 的等级	4906	20.98	
		3462	14.8	
	低于学位水平的	2577	11.02	
	高等教育：NVQ 4/NVQ5/ 学位或同等学历	4972	21.26	
	NA/无回答/拒绝	465	1.99	
经济活动	在职/学生[a]失业	13,097	97.32	
	者、退休人员或	627	2.68	
	其他非经济活跃人群	9664	41.32	
长期存在的限制性疾病	否[a]	17,567	75.11	
	是	5821	24.89	
婚姻状况	其他[a]已婚、同居或民事伴侣关系	8500	36.34	
		14,888	63.66	
儿童（2–15 岁）	没有一个	17,480	74.74	
	1	3013	12.88	
	≥2	2895	12.37	

（续见下一页）

Table 2 (*continued*)

Variable	(Category)	*n*/frequency	Mean/ % of total	*SD*
Infants (aged 0–2)	None	21,477	91.83	
	≥1	1911	8.17	
Car availability	No[a]	4521	19.33	
	Yes	18,851	80.6	
	NA/missing	16	0.07	
Survey year	2008[a]	15,098	64.55	
	2012	8290	35.45	

[a] Category used as the reference in the path model; variables without referents modelled as 'continuous'.
[b] Missing $n = 9$ (0.04%).
[c] Missing $n = 1914$ (8.18%).

survey' package (Oberski, 2014) in R version 3.3.3 (R Core Team, 2017). 'Lavaan survey' is an extension of the 'lavaan' package (Rosseel, 2012), designed for latent variable modelling, that calculates cluster-robust standard errors (SE) for a clustered sample (i.e. individuals within the same household) without specification of a multilevel model as such (McNeish et al., 2017). The main models are estimated with a maximum likelihood estimator with robust standard errors and a Satorra-Bentler scaled test statistic (MLM), robust to non-normality (Rosseel, 2012); an important consideration as outcomes and mediators were skewed (Supplementary Figs. A.1 and A.2). Because we used an estimator robust to non-normality, we did not transform any variables; this also provides coefficients which are more readily understandable. Although our health outcomes were measured in an ordinal scale (either 1–5 or 0–12), the use of linear estimation methods is justified as they have been found to be robust in large epidemiological samples (Lumley et al., 2002; Norman, 2010).

All 'not applicable' or 'refused' responses were recoded as missing (Table 2), with the exception of household income that showed a high number of missing values ($n = 4714$). In this case, to avoid systematic biases and to retain more data, all cases with income missing were recoded into 0, and we added a new binary variable in the models to indicate whether income was missing or not (Pedersen et al., 2017). As for coastal proximity, area-level deprivation and education, all variables that were ordered with unequally spaced categories, we tested whether including them in the model as 'continuous' or categorical variables (with the first category being the reference, see Table 2) would fit the data better (in terms of Akaike's and the Bayesian information criteria). In all cases, the categorical operationalisations fitted the data better and were subsequently adopted.

All environmental indicators, PA mediators, and health outcomes were included in the same analysis to account for their likely covariation. The error variances between outcomes as well as those between the PA variables were free to correlate. In the results, we were particularly interested in whether the indirect pathways from environmental indicators to the outcomes via outdoor PA statistically differed from 0 (that is, the estimate's $p < .05$ and 95% CI does not overlap with 0) using the delta method. We also conducted Wald's tests to compare the size of the indirect effects on the same outcomes.

2.4. Sensitivity analyses

To investigate whether walking, as the most common outdoor physical activity (Bélanger et al., 2011), might be the key on-land outdoor activity driving any mediation effects, additional analyses were run where on-land activities were separated into a) 'walking only' and b) 'all other' activities combined (namely, running, cycling and football/rugby), resulting in a model with four potential mediating pathways through PA (Sensitivity model 1).

Further sensitivity analyses were also run to make sure our models and main results were not merely due to specific model specifications. First, to ensure our approach to handling missing data for household income did not bias our main results, we re-estimated the models in two ways: excluding respondents whose income was unknown (resulting in $n = 17{,}195$; Sensitivity model 2a), and excluding income as a covariate altogether (Sensitivity model 2b). Second, as our health outcomes were not continuous (and skewed), we wanted to ensure our operationalisations did not affect the results and thus specified them as 'ordinal', and accordingly used a weighted least-square estimator (Sensitivity model 3). With this approach, however, the cluster-robust standard errors were not available (Rosseel, 2012). Third, we also examined all PA categories in terms of weekly hours conducted to see if it was the time spent being physically active, irrespective of intensity, that was associated with general and mental health (Sensitivity model 4). Fourth, to evaluate whether the environmental indicators could be related to the 'indoor/other PA' category due to e.g. less indoor facilities for PA in greener and possibly more rural settings, we specified a model with all paths from the environmental indicators to all PA variables (Sensitivity model 5). This type of model is saturated with 0 degrees of freedom, which means that the information on model fit is limited. Sixth, to evaluate the benefits of fitting a more comprehensive model, we specified and assessed single mediation models (with one explanatory variable, mediator and outcome at a time) corresponding to all indirect effects in the main model, both unadjusted and adjusted for the same set of covariates as the main model. Seventh, to assess any potential multicollinearity between blue and green space measures, we ran a model without coastal proximity and freshwater presence to examine if they confounded the effects of green space.

3. Results

3.1. Sample descriptives

On average, respondents rated their general and mental health in the past four weeks as good (Table 2; Supplementary Fig. A.1). In terms of PA patterns, in line with previous studies, the rates for watersports were relatively low in the sample (Papathanasopoulou et al., 2016; Elliott et al., 2018) whereas the majority of weekly MET hours were spent on on-land outdoor PA (Table 2; Supplementary Fig. A.2), mostly walking (mean 11.2 MET h/week, analysed in a separate model; see Section 3.4). The average PA rates, presented in Table 2, are approximately equivalent to 1 min (watersports), 151 min (on-land outdoor PA), and 75 min (indoor/other PA) of weekly cycling at a leisurely speed of 9.4 miles/h (equivalent to 5.8 METs/h; Ainsworth et al., 2011).

Regarding blue space indicators, the respondents most commonly lived > 50 km from the coast (38%), while less than a fifth lived within 5 km (17%, Table 2). Nearly a quarter had at least some freshwater in their area.

The intra-class correlations, reflecting the variance shared between the respondents in the same households, were substantial for all mediators and outcomes: 0.381 for general health, 0.221 for mental health, 0.118 for watersports, 0.230 for on-land outdoor PA, and 0.175 for indoor/other PA. The correlation coefficients between the outcomes and mediators are provided as a Supplementary material (Table A.1).

3.2. Model fit and variances explained

The model showed excellent fit with the data: χ^2 (robust) = 0.059, df = 6, p = 1.000; CFI = 1.00; TLI = 1.06; RMSEA < 0.001; SRMR = 0.001. The largest normalised residuals were within acceptable limits (Kline, 2016); 1.23 and 0.89, and all others were < |0.34|. Modification indices showed no apparent missing paths. The variances explained in the outcomes were 36.2% for general health and 12.2% for mental health. The model explained a smaller, yet still substantial share of variation in outdoor PA on land (5.2%) and other PA (6.8%), whereas outdoor PA on water was poorly explained (0.3%; Ferguson,

表 2（续）

变量	（类别）	n/频率	平均值/占总值的百分比	*SD*
婴儿（0–2 岁）	没有一个	21,477	91.83	
	≥1	1911	8.17	
车辆供应情况	否[a]	4521	19.33	
	是	18,851	80.6	
	NA/缺失	16	0.07	
调查年份	2008[a]	15,098	64.55	
	2012	8290	35.45	

[a] 在路径模型中用作参考的类别；无参照变量的建模方式为“ζ continuous′ ”。

[b] 缺失例数 n = 9（0.04%）。

[c] 缺失病例数 n = 1914（8.18%）。

使用 R 语言 3.3.3 版本中的 `survey` 包（Oberski，2014）（R Core Team，2017）。`ζLavaan survey` 是 `ζlavaan` 包（Rosseel，2012）的扩展功能，专为潜变量建模设计，可在无需明确指定多层模型的情况下，为聚类样本（即同一家庭中的个体）计算聚类稳健标准误（SE）（McNeish et al.，2017）。主要模型采用具有稳健标准误的最大似然估计量及 Satorra-Bentler 缩放检验统计量（MLM）进行估计，该统计量对非正态性具有稳健性（Rosseel，2012）；鉴于结局变量与中介变量均呈偏态分布（见补充图 A.1 与 A.2），此考量至关重要。由于采用了对非正态性稳健的估计量，我们未对任何变量进行转换；此举亦使所得系数更易被理解。尽管健康结局变量采用序数尺度测量（分别为 1–5 或 0–12），仍可采用线性估计方法，因为已有研究证实该方法在大型流行病学样本中具有稳健性（Lumley et al.，2002）。Norman，2010)。

所有标记为“ζ不适用”或“ζ拒绝”的应答均被重新编码为缺失值（表2），但家庭收入变量除外——该变量存在大量缺失值（n = 4714）。针对此情况，为避免系统性偏差并保留更多数据，所有收入缺失的观测值均被重新编码为0，并在模型中新增一个二元变量以指示收入是否缺失（Pedersen 等，2017）。对于沿海邻近性、区域贫困程度及教育水平等采用非等距分类变量的指标，我们分别测试将其作为“ζ连续型”变量或分类变量（以首类为参照）纳入模型时，哪种方式能更好地拟合数据（基于赤池信息量准则与贝叶斯信息准则）。在所有案例中，分类变量操作化方式均能更好地拟合数据，故最终予以采纳。

为考虑各环境指标、身体活动（PA）中介变量及健康结局之间可能存在的共变关系，本研究将上述所有变量纳入同一分析框架。结局变量之间以及各PA变量之间的误差方差均允许存在相关性。在结果分析中，我们特别关注：通过δ法检验，环境指标经由户外PA传导至结局变量的间接效应路径是否在统计学上显著不同于0（即估计值的p值<0.05，且95%置信区间与0无重叠）。此外，我们还采用Wald检验比较相同结局变量上各间接效应的效应量。

2.4. 敏感性分析

为探究步行——作为最常见的户外体力活动（Belanger 等，2011）——是否可能是驱动任何中介效应的关键陆地户外活动，我们进行了额外分析：将陆地活动分为 a) “仅步行”和 b) “其他所有活动”（即跑步、骑行及足球/橄榄球）的组合，从而构建出包含四条潜在体力活动中介路径的模型（敏感性模型 1）。

我们还进行了进一步的敏感性分析，以确保我们的模型及主要结果并非仅源于特定模型设定。首先，为确保我们对家庭收入缺失数据的处理方法不会对主要结果产生偏倚，我们采用两种方式重新估计模型：一是排除收入未知的受访者（此时样本量 n = 17,195；敏感性模型 2a），二是完全排除收入作为协变量（敏感性模型 2b）。其次，由于我们的健康结局变量并非连续变量（且呈偏态分布），我们力求确保变量操作化方式不会影响结果，因此将其设定为“ζ序数型”，并相应地采用加权最小二乘估计量（敏感性模型 3）。然而，采用此方法后，无法获得聚类稳健标准误（Rosseel，2012）。第三，我们还针对所有体力活动类别，考察每周进行的体力活动时长——无论强度如何——是否与整体及心理健康相关（敏感性模型 4）。第四，为评估环境指标是否可能与“室内/其他体力活动”类别相关（例如由于.....）。鉴于在更绿色且可能更具乡村特色的环境中，用于身体活动（PA）的室内设施较少，我们构建了一个包含从环境指标到所有身体活动变量的全部路径的模型（敏感性模型5）。此类模型具有0个自由度，这意味着模型拟合度的信息量有限。第六，为评估构建更综合模型的优势，我们针对主模型中所有间接效应，分别构建并评估了单一中介模型（每次仅包含一个解释变量、一个中介变量和一个结局变量），包括未调整模型以及使用与主模型相同协变量集进行调整的模型。第七，为评估蓝色空间与绿色空间测量指标之间是否存在潜在多重共线性，我们运行了一个不含沿海邻近性与淡水分布变量的模型，以检验这两项变量是否会对绿色空间的效应产生混杂影响。

3. 结果

3.1. 样本描述性统计

平均而言，受访者对过去四周内的总体及心理健康状况评价为良好（表2；补充图A.1）。在体力活动模式方面，与既往研究一致，样本中水上运动参与率相对较低（Papathanasopoulou 等，2016；Elliott 等，2018），而每周MET时数中大部分用于陆地户外体力活动（表2；补充图A.2），其中主要为步行（平均每周11.2 MET·小时，已通过独立模型分析；参见第3.4节）。表2中呈现的平均体力活动时长，约相当于以9.4英里/小时（相当于5.8 METs/小时；Ainsworth等，2011）的休闲速度每周进行1分钟（水上运动）、151分钟（陆地户外体力活动）及75分钟（室内/其他体力活动）的骑行时间。

关于蓝色空间指标，受访者中最常见的居住地点距海岸线超过 50 公里（38%），而居住在 5 公里范围内的受访者比例不足五分之一（17%，表 2）。近四分之一的受访者所在地区至少拥有一定规模的淡水资源。

类内相关系数反映了同一家庭内受访者间共享的方差，所有中介变量与结局指标均呈现显著相关性：总体健康为0.381，心理健康为0.221，水上运动为0.118，陆地户外体力活动为0.230，室内/其他体力活动为0.175。结局指标与中介变量之间的相关系数已作为补充材料（表A.1）提供。

3.2. 模型拟合度及解释的方差

该模型与数据展现出极佳拟合度：χ^2（稳健）= 0.059，df = 6，p = 1.000；CFI = 1.00； TLI = 1.06； RMSEA < 0.001； SRMR = 0.001。最大标准化残差均在可接受范围内（Kline，2016）：分别为1.23与0.89，其余残差均满足< |0.34|>。修正指数未显示明显缺失路径。模型对结局变量的解释方差分别为：总体健康36.2%，心理健康12.2%。该模型对陆地户外体力活动（5.2%）及其他户外体力活动（6.8%）变异的解释比例

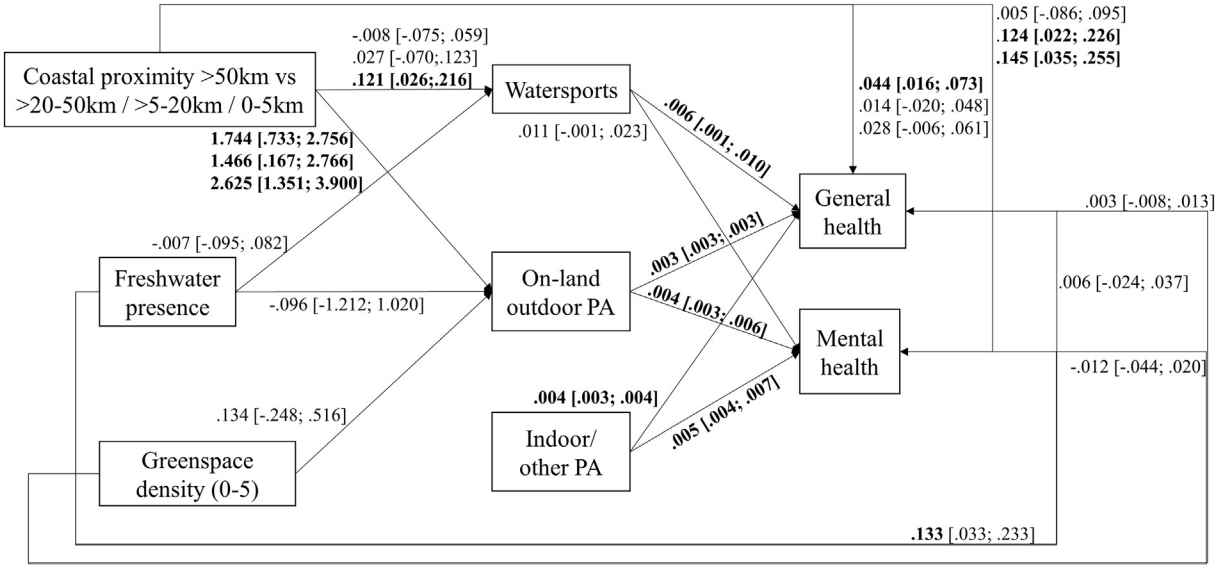


Fig. 2. Unstandardised estimates and their 95% CIs in the path model (n = 21,097), adjusted for all covariates (provided in Supplementary Table B.1). All PA variables are measured in MET h/week. Error covariances between the mediators and the outcomes are not shown for readability but they are provided in Supplementary Table B.1. Bold typeface indicates statistical significance.

2009). We suspect this is due to overall low levels of watersports in general (Table 2).

3.3. Model estimates

Compared to people who lived > 50 km from the coast, those who lived in all distance bands closer to the coast reported more on-land outdoor PA, with the effect strongest for those living < 5 km (Fig. 2, Table 3). On-land outdoor PA in turn was associated with significantly better general and mental health. These connections were strong enough to establish significant indirect connections from coastal proximity to general and mental health via on-land outdoor PA (Table 3). Wald's tests indicated that these indirect effects were approximately equal. Those who lived < 5 km from the coast reported more watersports than those living > 50 km from the coast (b = 0.121, s.e. = 0.048) but the indirect effects on general and mental health were small and not statistically significant (Table 3). Mental health was 0.124 and 0.145 units greater (on the 0–12 scale) for those living 5 km–20 km and < 5 km from the coast, respectively (compared with > 50 km), whereas general health was 0.044 units greater (on the 1–5 scale) for those living 20–50 km from the coast, compared to > 50 km (Fig. 2, Table 3). Overall, the strongest total effect between coastal proximity and outcomes was that from coastal proximity < 5 km versus > 50 km on mental health (Table 3).

Freshwater presence was directly connected to better mental health (b = 0.133, s.e. = 0.051) but there was no significant association with on-land outdoor PA or watersports (Fig. 2, Table 3). Contrary to expectations, green space density was associated with neither outdoor PA, general health nor mental health (Fig. 2, Table 3).

All covariates, except rural/urban status, showed significant associations with at least one of the outcomes or the mediators (Supplementary Table B.1). The associations for the socio-economic covariates were mostly in the expected directions (for example, more educated individuals were generally healthier and conducted more PA). Regarding the correlates of PA, car ownership was associated with less on-land outdoor PA but more for indoor/other PA.

3.4. Sensitivity analyses

Results for our first sensitivity analysis, which disaggregated all on-land outdoor PA into walking vs. other (i.e. running, cycling, football/rugby) can be seen in Appendix C. This model also showed good fit, although the variance explained for walking itself was relatively low (1.9%) in comparison to all other on-land outdoor activities (7.6%; Supplementary Tables C.1 & C.2). Most importantly, the associations between coastal proximity and walking were significant and similar -although weaker- to those in the main model regarding all outdoor on-land PA, whereas the associations between coastal proximity and the

Table 3
Total, direct, and indirect effects (estimates and 95% CIs) from blue/green space indicators to general and mental health via watersports and on-land outdoor PA. In bold: p < .05. If the estimate is | < 0.001|, no CI shown for clarity.

Env. indicator	Health outcome	Total effect	Direct effect	Indirect effects via	
				Watersports	On-land outdoor PA
Coastal proximity (20–50 km vs. > 50 km)	General	**0.049 [0.021;0.078]**	**0.044 [0.016;0.073]**	< |0.001|	**0.005 [0.002;0.008]**
	Mental	0.012 [−0.078;0.102]	0.005 [−0.086;0.095]	< |0.001|	**0.008 [0.003;0.012]**
Coastal proximity (5–20 km vs. > 50 km)	General	0.019 [−0.015;0.053]	0.014 [−0.020;0.048]	< |0.001|	**0.004 [0.001;0.008]**
	Mental	**0.130 [0.028;0.232]**	**0.124 [0.022;0.226]**	< |0.001|	**0.006 [< |0.001|;0.012]**
Coastal proximity (0–5 km vs. > 50 km)	General	**0.036 [0.003;0.070]**	0.028 [−0.006;0.061]	0.001 [< |0.001|;0.001]	**0.008 [0.004;0.012]**
	Mental	**0.157 [0.048;0.267]**	**0.145 [0.035;0.255]**	0.001 [< |0.001|;0.003]	**0.011 [0.005;0.018]**
Freshwater presence	General	0.006 [−0.024;0.037]	0.006 [−0.024;0.037]	< |0.001|	< |0.001|
	Mental	**0.132 [0.032;0.232]**	**0.133 [0.033;0.233]**	< |0.001|	< |0.001|
Green space density	General	0.003 [−0.007;0.013]	0.003 [−0.008;0.013]	na	< |0.001|
	Mental	−0.011 [−0.043;0.021]	−0.012 [−0.044;0.020]	na	0.001 [−0.001;0.002]

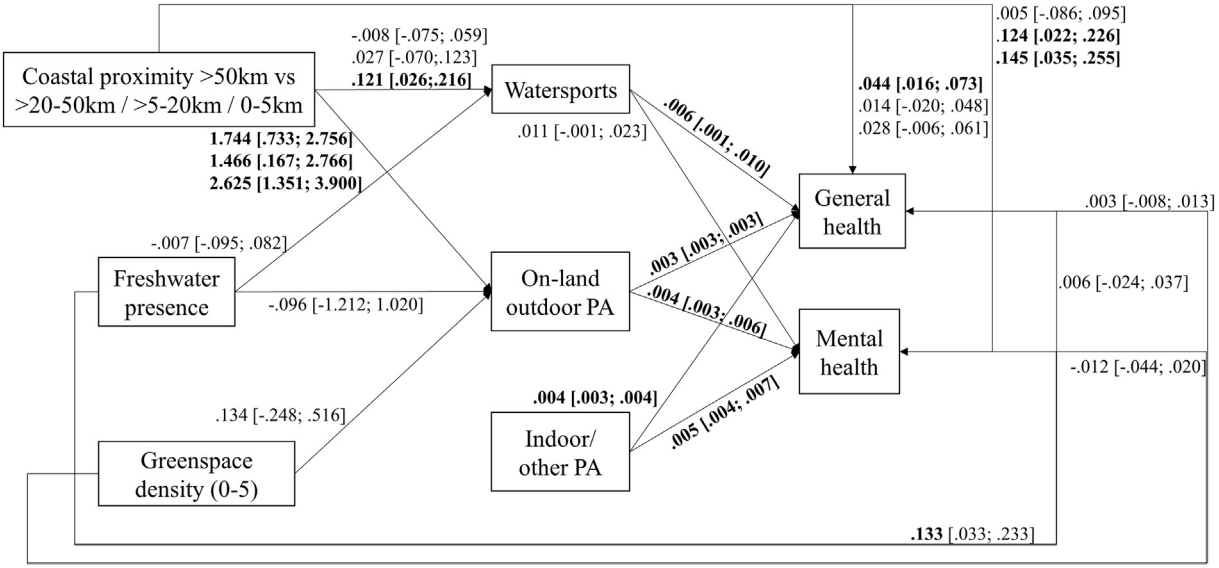


图 2. 路径模型中的未标准化估计值及其 95% 置信区间（n = 21,097），已针对所有协变量进行调整（详见补充表 B.1）。所有 PA 变量均以“METh/week”为单位测量。为保持版面整洁，中介变量与结局变量之间的误差协方差未予显示，但已收录于补充表 B.1 中。粗体标示具有统计学显著性。

较小，但对水上户外体力活动的解释度较低（0.3%；Ferguson，2009）。我们认为这可能与水上运动总体参与率较低有关（表2）。

3.3. 模型估计值

与距海岸线>50公里的人群相比，生活在所有距海岸线较近距离区段的人群报告了更多陆地户外体力活动，且这种效应对于距海岸线<5公里的人群最为显著（图2，表3）。而陆地户外体力活动又与显著改善的整体健康及心理健康相关。这些关联强度足以通过陆地户外体力活动，建立从海岸线邻近性到整体健康及心理健康的显著间接效应路径（表3）。Wald检验表明，这些间接效应大小近似相等。距海岸线<5公里的人群比距海岸线>50公里的人群报告更多水上运动参与（b=0.121，s.e.= 0.048），但对整体健康及心理健康的间接效应较小且无统计学显著性（表3）。与距海岸线>50公里的人群相比，距海岸线5–20公里及<5公里人群的心理健康评分分别高出0.124与0.145个单位（采用0–12分量表）；而距海岸线20–50公里人群的整体健康评分较>50公里人群高出0.044个单位（采用1–5分量表）（图2，表3）。总体而言，沿海距离与结局指标之间最显著的总体效应，体现在沿海距离 < 5 km 与 > 50 km 对心理健康的影响上（表 3）。

淡水资源的存在与更好的心理健康状况直接相关（b = 0.133，s.e. = 0.051），但与陆地户外体力活动或水上运动并无显著关联（图2，表3）。与预期相反，绿地密度既与户外体力活动、总体健康状况亦与心理健康状况均无显著关联（图2，表3）。

除城乡居住状况外，所有协变量均与至少一项结局指标或中介变量存在显著关联（补充表 B.1）。社会经济协变量的关联方向大多符合预期（例如：受教育程度较高者通常健康状况更佳，且进行更多体力活动）。关于体力活动的相关因素，汽车拥有量与陆地户外体力活动减少相关，但与室内/其他形式体力活动增加相关。

3.4. 敏感性分析

我们在首次敏感性分析中，将所有陆地户外体力活动细分为步行与其他类型（如跑步、骑行、足球/橄榄球）的结果详见附录C。该模型同样显示出良好的拟合度，但相较于其他所有陆地户外活动（7.6%；补充表C.1与C.2），步行本身的方差解释率相对较低（1.9%）。最关键的是，沿海邻近性与步行之间的关联在敏感性分析中仍具统计学显著性，且与主模型中所有陆地户外体力活动的相关性趋势相似——尽管关联强度稍弱；而沿海邻近性与

表 3
通过水上运动及陆地户外体力活动，从蓝/绿地指标对总体健康与心理健康产生的总效应、直接效应及间接效应（估计值及 95% 置信区间）。加粗部分表示：p < .05。若估计值 | < 0.001|，为清晰起见，未显示置信区间。

环境指标	健康结局	总计及影响	直接效应	通过以下途径产生的间接影响	
				水上运动	陆地户外体力活动
距海岸线距离（20–50 km vs. > 50 km）	概要	**0.049 [0.021;0.078]** 0.012 [−0.078;0.102] 0.019 [−0.015;0.053] **0.130 [0.028; 0.232] 0.036 [0.003; 0.070] 0.157 [0.048; 0.267]** 0.006 [−0.024; 0.037] **0.132 [0.032;0.232]** 0.003 [−0.007;0.013] −0.011 [−0.043;0.021]	**0.044 [0.016;0.073]** 0.005 [−0.086;0.095] 0.014 [−0.020;0.048] **0.124 [0.022;0.226]** 0.028 [−0.006;0.061] **0.145 [0.035;0.255]** 0.006 [−0.024;0.037] **0.133 [0.033;0.233]** 0.003 [−0.008;0.013] −0.012 [−0.044;0.020]	< |0.001| < |0.001| < |0.001| < |0.001| 0.001 [< |0.001|;0.001] 0.001 [< |0.001|;0.003] < |0.001| < |0.001| na na	**0.005 [0.002;0.008] 0.008 [0.003;0.012] 0.004 [0.001;0.008] 0.006 [** < |0.001|;0.012] **0.008 [0.004;0.012] 0.011 [0.005;0.018]** < | 0.001| < |0.001| < |0.001| 0.001 [−0.001;0.002]
距海岸线距离（5–20 km vs. > 50 km）	心理				
距海岸线距离（0–5 km vs. > 50 km）| 淡水	概要				
分布情况	心理				
绿地密度	概要				
	心理				

remaining on-land outdoor physical activities were much weaker and not significant (Supplementary Table C.1, Fig. C.1). Walking also showed a positive indirect effect from coastal proximity to general and mental health (Supplementary Table C.2). These results suggest that it was indeed walking, rather than other forms of on-land outdoor PA which may be mostly responsible for the mediation pattern we see between coastal proximity and general and mental health in the main model.

In addition, Sensitivity model 1 indicated that some of the covariates were differently associated with walking and other on-land outdoor PA. For example, gender, age, and income were either weakly or not associated with walking but significantly explained other on-land outdoor PA (Supplementary Table C.1). Regarding education, the reverse was true so that the more educated walked more but did not conduct more of other types of on-land outdoor PA, compared with those with less formal education.

Main results regarding the associations between blue spaces, outdoor PA, and the health outcomes were replicated in most sensitivity analyses, adding confidence to our findings. In Sensitivity models 2a (Appendix Table D.1) and 3, the estimates between coastal proximity and watersports were, however, smaller than in the main model (b = 0.090–0.010, s.e. = 0.046–0.060) and not statistically significant; in addition, both models showed a positive but weak connection between green space density and general health (b = 0.013, s.e. = 0.005–0.006) but no connection between 5 and 20 km distance to the coast and mental health. In Sensitivity model 3, coastal proximity < 5 km was also weakly but significantly connected to general health (b = 0.052, s.e. = 0.022). In Sensitivity model 4, measuring PA in weekly hours (Appendix Table E.1), the relationship between coastal proximity 5–20 km (versus > 50 km) and on-land outdoor PA was positive but not statistically significant. Sensitivity model 5 (Appendix Table F.1), with all possible paths assessed, did not substantially differ from our main model.

The sixth set of sensitivity analyses assessed each indirect relationship separately. The models that were unadjusted for the covariates replicated the results regarding the associations between the environmental indicators and outdoor PA, whereas there were some inconsistencies in the direct associations between environmental indicators and the outcomes. Unexpectedly, many of the adjusted single mediation models had issues with convergence. These seemed to have been caused by one/some of the covariates but we could not identify a single one that would have solved these issues in all models. Hence, in the case of this study, assessing associated mediation mechanisms between environmental exposures and health outcomes in parallel (that is, in a single model) was not only the recommended (Markevych et al., 2017) but also a necessary approach to be able to adjust the model for a large number of a priori chosen potential confounders.

Finally, the model without blue space indicators (Sensitivity model 7), showed a small effect between green space and general health (b = 0.009, s.e. = 0.004, p = .026) but no associations with mental health or any of the mediators, implying that the lack of significant associations between green space and PA/outcomes was not merely due to multicollinearity.

4. Discussion

4.1. Main findings

Supporting previous work globally (Gascon et al., 2017), the current study found that coastal respondents in England reported: a) better general health (e.g. Wheeler et al., 2012); b) better mental health (e.g. White et al., 2013a); and c) more recreational PA (e.g. White et al., 2014), than those living inland. Extending previous findings, we were able to: a) disaggregate recreational PA into different types (e.g. watersports, outdoor on-land, and indoor); and b) show that it was the amount of on-land outdoor recreational activity (especially walking), as opposed to watersports, which mediated the relationships between coastal proximity and both general and mental health. Put simply, adults in England who reside in coastal areas tend to be happier and healthier than similar individuals inland, in part because they engage in more outdoor physical activity such as walking.

The relationships were meaningful in terms of scale. For instance, the difference in on-land outdoor PA between living < 5 km from the coast compared with > 50 km away was equivalent, for example, to cycling 14–40 min more per week at 9.4 mph (equivalent of 5.8 METs/h; Ainsworth et al., 2011). These relationships held in a series of sensitivity analyses. Our findings suggest that a part of the positive association between coastal proximity and mental health, established in previous research (e.g. Gascon et al., 2017) can be explained by PA conducted outdoors on land, and thus highlights the importance of capacity building (Markevych et al., 2017) as one of the mechanisms behind this association. Yet, there remained a substantial share of the variance in our outcomes that was not related to PA, which could be due to the other established mechanisms such as social cohesion and stress reduction (Gascon et al., 2015). Furthermore, while it may be plausible to assume that those living the closest to the coast may use coastal settings for their PA routines (e.g. White et al., 2014), we can only speculate if this is true for residents of areas 5–50 km from the coast who also showed increased amounts of outdoor PA compared with those living more inland. It could be that in these areas not in immediate vicinity of the coast, other amenities account for increased on-land PA. Alternatively, residents in these areas may use the coast for PA less frequently but compensate by staying longer and consuming more energy per visit (Elliott et al., 2015). Further research is needed to investigate these possibilities.

Walking, apart from other on-land outdoor PA, partially mediated the relationship between residential coastal proximity and both general and mental health. Respondents living within 5 km of the coast on average undertook 20–56 min more of weekly, moderate-intensity walking (equivalent of 3.5 METs/h; Ainsworth et al., 2011) compared to those further inland. In line with previous work, this could indicate that walking is a popular activity to do at coastal environments (Elliott et al., 2018) that is sufficiently health-enhancing (Elliott et al., 2015), but we cannot definitively state that the actual coastline supported walking, as it may be that coastal towns and cities simply have better infrastructure for walking. Nonetheless, the development of England's fully publicly accessible coastline (Natural England, 2013) could indicate that areas in very close proximity of the sea are used for health-enhancing physical activity. This finding also has important public health implications. Previous research has demonstrated that walking, separately from all other forms of PA, can protect against physical health challenges like cardiovascular disease (Hamer and Chida, 2008) and mental health difficulties like depression (Robertson et al., 2012), and thus coastal environments could reduce the national burden of such diseases through supporting PA.

We also found evidence of a positive connection between residential freshwater coverage and mental health. This association was, in effect size, comparable to coastal proximity (< 5 km vs. > 50 km). Contrary to coastal proximity, however, the positive connection between freshwater presence and mental health was not mediated by any type of outdoor PA. It remains a topic for future studies to examine other mechanisms that link freshwater presence and mental health. Tentative evidence suggests that this connection could be better explained by perceived psychological benefits and social interaction (De Bell et al., 2017). Alternatively, the recreational use of freshwater settings in England may be limited by their small size and issues with quality and amenities such as cleared pathways. For example, some estuarine areas are counted in the freshwater measures but they might not be suitable for recreational use due, for instance, to varying levels of water coverage.

Green space was not significantly associated with general or mental health, or any PA type. This result contrasts with many previous

其余陆地户外体力活动的影响则显著减弱且无统计学意义（补充表C.1，图 C.1）。步行对整体健康及心理健康亦表现出正向间接效应（补充表 C.2）。这些结果表明，在主模型中，沿海邻近性与整体及心理健康之间的中介效应模式，确实主要由步行活动而非其他陆地户外体力活动所驱动。

此外，敏感性模型1表明，部分协变量与步行及其他陆地户外体力活动（PA）的关联模式存在差异。例如，性别、年龄及收入变量与步行的关联较弱或无关联，但对其他陆地户外体力活动具有显著解释力（补充表C.1）。而在教育水平方面，情况则相反：与受教育程度较低者相比，受教育程度较高者步行频率更高，但从事其他类型陆地户外体力活动的频率并未增加。

关于蓝色空间、户外体力活动（PA）与健康结局之间关联的主要研究结果，在大多数敏感性分析中均得到复现，这进一步增强了我们研究结论的可信度。在敏感性模型2a（附录表D.1）与模型3中，沿海邻近性与水上运动之间的效应量虽较主模型有所减小（b = 0.090–0.010，s.e. = 0.046–0.060），但均无统计学显著性；此外，这两个模型均显示绿地密度与总体健康呈正向但微弱关联（b = 0.013，s.e. = 0.005–0.006），而距海岸线5 km与20 km范围内的绿地密度与心理健康则无显著关联。在敏感性模型3中，距海岸线<5 km的区域与总体健康亦存在微弱但显著的关联（b = 0.052，s.e. = 0.022）。在敏感性模型4中（以每周体力活动时长为测量指标，见附录表E.1），距海岸线5–20 km（相较于>50 km）区域与陆地户外体力活动之间的关联呈正向，但无统计学显著性。包含所有可能路径的敏感性模型5（见附录表F.1）结果与主模型相比并无实质性差异。

第六组敏感性分析分别对每项间接关联进行了评估。未对协变量进行调整的模型均复现了环境指标与户外体力活动之间关联性的研究结果，但在环境指标与结局指标之间的直接关联方面，部分模型结果存在不一致性。出乎意料的是，许多经调整的单一中介模型均出现收敛性问题。这些问题似乎由某一或若干协变量引起，但我们未能确定一个可在所有模型中解决此类问题的单一协变量。因此，在本研究中，并行评估环境暴露与健康结局之间的中介机制（即在单一模型中完成）不仅符合推荐建议（Markhevych等，2017），更是针对大量预先选定的潜在混杂因素进行模型调整的必要方法。

最后，未纳入蓝色空间指标的模型（敏感性模型7）显示，绿色空间与总体健康之间存在微弱关联（b = 0.009，s.e. = 0.004，p = 0.026），但与心理健康或任何中介变量均无显著关联；这表明绿色空间与体力活动/结局指标之间缺乏显著关联，并非单纯由多重共线性所致。

4. 讨论

4.1. 主要研究结果

为支持既往全球性研究（Gascon 等，2017），本研究发现英格兰沿海地区的受访者相较于内陆居民报告：a) 总体健康状况更佳（如Wheeler等，2012）；b) 心理健康状况更佳（如White等，2013a）；以及 c) 更多休闲体力活动（如White等，2014）。基于既往研究结果，我们进一步实现了以下目标：a) 将休闲体力活动细分为不同类型（如水上运动、户外陆地活动及室内活动）；b) 证明正是陆地户外休闲活动（尤其是步行）——而非水上运动——在沿海居住与总体及心理健康之间起到了中介作用。简言之，英格兰沿海地区的成年人往往比内陆地区的同类人群更快乐、更健康，部分原因在于他们参与了更多户外体力活动，例如步行。

这些关系在量级上具有重要意义。例如，与距海岸线>50公里的人群相比，距海岸线<5公里的人群在陆地户外体力活动方面的差异，相当于每周多骑行14–40分钟、时速9.4英里（相当于5.8 METs/h；Ainsworth等，2011）。这些关系在一系列敏感性分析中均成立。我们的研究结果表明，先前研究（如Gascon等，2017）中确立的沿海居住与心理健康之间的正向关联，部分可由陆地户外体力活动来解释，从而凸显了能力建设（Markevych 等，2017）作为该关联背后机制之一的重要性。然而，我们的研究结果中仍有相当一部分变异与体力活动无关，这或许可归因于其他已确立的机制，如社会凝聚力与压力缓解（Gascon 等，2015）。此外，尽管可以合理推测距海岸线最近的居民可能将沿海环境纳入其日常体力活动模式（如White等，2014），但我们仅能进行推测。对于距离海岸线5–50公里的居民而言，这一结论同样成立——与居住在内陆地区的居民相比，这些居民的户外体力活动量也有所增加。在这些并非紧邻海岸线的地区，其他便利设施可能是导致陆地体力活动量增加的原因。或者，这些地区的居民可能较少利用海岸线进行体力活动，但通过延长每次外出停留时间并增加每次外出时的能量消耗来弥补这一不足（Elliott et al.，2015）。仍需进一步研究以探究这些可能性。

步行与其他陆地户外体力活动一样，部分介导了居住区沿海邻近性与总体及心理健康之间的关系。居住在距海岸线5公里范围内的受访者，每周进行中等强度步行（相当于3.5 METs/h；Ainsworth等，2011）的时间平均比居住在内陆地区的受访者多出20–56分钟。与既往研究一致，这可能表明步行是沿海环境中广受欢迎的活动（Elliott等，2018），且足以促进健康（Elliott 等，2015）；但我们无法断言实际海岸线本身支持步行活动，因为沿海城镇和城市可能仅拥有更完善的步行基础设施。尽管如此，英格兰已建成完全向公众开放的海岸线（Natural England，2013），这可能表明紧邻海洋的区域已被用于促进健康的体力活动。这一发现亦具有重要的公共卫生意义。既往研究表明，与所有其他形式的体力活动不同，步行本身就能预防心血管疾病等身体健康问题（Hamer 与 Chida，2008），以及抑郁症等心理健康问题（Robertson 等，2012）；因此，沿海环境可通过促进体力活动，减轻此类疾病对国家的负担。

我们还发现了居住区淡水覆盖面积与心理健康之间存在正向关联的证据。这种关联的效应量与沿海邻近性（<5 km 与 >50 km）相当。然而与沿海邻近性不同，淡水存在与心理健康之间的正向关联并未通过任何类型的户外体力活动介导。未来研究仍需探索其他将淡水存在与心理健康相联系的机制。初步证据表明，这种关联或许更可通过个体感知的心理益处及社交互动来解释（De Bell 等人，2017）。另一种可能是，英格兰地区淡水环境的休闲利用可能受限于其狭小空间以及水质与配套设施（如已清理的步道）方面的不足。例如，某些河口区域虽被纳入淡水统计范畴，却可能因水域覆盖程度不一等因素而不适合休闲活动。

绿地与总体健康或心理健康，以及任何类型的体力活动均无显著关联。这一结果与许多既往研究相悖。一种可能的解释是，居住区的蓝色

studies. One possibility might be that residential blue space is more strongly connected to mental health than green space (Nutsford et al., 2016; de Vries et al., 2016), however far more research is needed before such a conclusion can be made, not least greater clarity about the types of, and access to, different sorts of green and blue spaces in different residential areas. Another plausible explanation for these null findings could be that we had no available measure of perceived green space or its quality. Recent evidence suggests that greenery visible on the street level may be more relevant for mental health than greenery measured from the aerial perspective (Helbich et al., 2019). Relatedly, another possibility for the lack of association between area-level greenspace and health outcomes is the greater measurement error in how green space is assessed. While we can be reasonably confident that Euclidean distances to coasts are measured accurately, percentages of green space are dependent on a number of other assumptions about what constitutes "green space" and how and when land use maps are produced (Helbich, 2019). For example, the presence of street trees has been show to correlate with indicators of physical (de Vries et al., 2013) and mental (Taylor et al., 2015) health previously, but could not be accounted for in the present study. Thus, here we potentially underestimated exposure to green space. Furthermore, quality issues, such as facilities, connectivity, cleared pathways, and restorative perceptions may also be more relevant than density in terms of how green spaces are used (Dzhambov et al., 2018; Francis et al., 2012). The relationship between green space and PA may also vary between different green space indicators and buffers (Klompmaker et al., 2018). Options for capturing better the environments visited in everyday life for PA include dynamic exposure assessments (Helbich, 2018) and assessing larger areas or buffers around one's residence (Browning and Lee, 2017). However, in the case of our study, using a larger surrounding area may have resulted in the same conclusions. This was indicated by Mytton et al. (2012) who used earlier waves of the HSE with green space measured on the geographically larger middle super output area (MSOA) scale (as opposed to LSOAs used here), and found no association between green space and outdoor PA either. Although systematic reviews have often concluded that there seems to be a positive but small connection between green space and outdoor PA, there are a large number of studies that also report no such relationship (van den Bosch and Ode Sang, 2017; Klompmaker et al., 2018). Finally, recent research using English population data indicates that dog ownership may be an important moderator of the association between neighbourhood green space density and PA (White et al., 2018); since the HSE does not include information on dog ownership in the same years as physical activity we could not investigate this here, but it may have confounded our findings.

4.2. Strengths, limitations and further studies

Our paper makes several contributions to the field. First, we address the role of physical activity, one of the major pathways that links residential blue and green spaces to health and well-being, in greater detail than many previous studies in this topic (Markevych et al., 2017). By separating physical activity into watersports, outdoors on-land, or in indoor/other, we can better understand how residential neighbourhood environments and their use are associated with general and mental health (Markevych et al., 2017). Second, we have a large epidemiological dataset with standard, reliable measures of general and mental health. The survey data permitted adjustment of analyses for a large number of potential confounders addressed in past research without compromising on statistical power, and our results on these variables are comparable with other studies (van den Berg et al., 2015; Gascon et al., 2017; Markevych et al., 2017). Third, by using path modelling as our analytical approach, we assessed the interrelationships between residential blue and green space, PA in different types of settings, and general and mental health in a single model (as recommended by Markevych et al., 2017). This way potential confounding between the mediators, outcomes, and explanatory variables is better accounted for than in the case of more simplistic models that analyse these variables separately. However, we did not assess whether some of the covariates, such as socio-economic factors (Wheeler et al., 2012), moderate the associations between coastal proximity and PA or the outcomes; these were outside the scope of this study but worthy of investigation in the future. Fourth, although outdoor PA, especially walking, seems tentatively more frequent near blue spaces than further from them (e.g. Humpel et al., 2004), to our knowledge no previous epidemiological studies have assessed whether this relationship is strong enough to associate with better perceived general or mental health (Gascon et al., 2017).

However, this study is was not without limitations, and many of these are typical for studies based on cross-sectional survey data. This study cannot imply causality, of which evidence is generally limited (Gascon et al., 2015). Based on our analysis we can say that those who live closer to the coast conduct more outdoor PA (especially walking) and feel healthier than those living > 50 km away from the coast. However, we cannot say whether more active and healthier people tend to move closer to the coast or if it is coastal proximity that encourages people to be more physically active and, consequently, healthier. Moreover, we relied on self-reported PA in the current analyses when there is evidence that these may be exaggerated (Hagstromer et al., 2010). Although a sub-sample of the 2008 HSE participants did wear accelerometers for a period during the study, we were unable to use this data here because it was not linked to information on the activities or geospatial factors. Even if we had have been able to use it here, neither self-report nor objectively-measured PA is perfect, of course: while self-reports tend to be exaggerated, the overestimations are relatively similar across different demographic groups and activities, whereas accelerometry is unable to capture some common activities such as cycling and, of potentially crucial importance for the current work, swimming and other watersports (Aresu et al., 2009a).

The current work was also limited by relatively coarse environmental data being appended by the data provider which was necessary to ensure respondent anonymity. For this reason we were not able to assess any regional or seasonal patterns (Scholes and Mindell, 2013), or to measure green space or freshwater elements in larger areas surrounding the respondents' homes (e.g. by identifying neighbouring areas or by calculating these indicators within a set distance), which may be more relevant in terms of PA behaviour (Giles-Corti et al., 2005). Being restricted to analysing all environmental information at the LSOA level prevented us from assessing the modifiable area unit problem, that is, if using other administrative boundaries or buffers would have yielded different results (although we do find comparable results to those of Mytton et al. (2012) on greenspace and PA using larger spatial units as described above). Similarly, we were unable to obtain the exact values for our environmental indicators (such as exact distance to the coast or green space density) but could only use broad categories that prevented more refined analysis.

Finally, in terms of limitations, although we had precise information on the type of PA conducted, we could not link those activities specifically to a certain setting or a type of setting, a research priority stated by Markevych et al. (2017). Thus, our PA categories were based solely on activities and were quite broad. Some activities classified as 'indoor/other PA' could have been conducted outdoors. In the sensitivity analysis (Appendix Table F.1) where we assessed the relationship from the environmental indicators to 'indoor/other PA', none of the associations were significant, potentially justifying our assumption that the major forms of outdoor PA were captured by the 'on-land outdoor PA' measure (Mytton et al., 2012). Regarding watersports, the biggest drawback was that we could not reliably include swimming in the measure since there was no information on whether swimming was conducted indoors or outdoors. This may partly explain why watersports, as a group of activities, were relatively uncommon (although the result is consistent with other surveys; Elliott et al., 2018); its variance explained was low,

空间与心理健康的关联可能比绿地更为密切（Nutsford等人，2016；de Vries等人，2016），但在此结论得出之前仍需开展更多研究——尤其是需要更清晰地了解不同居住区中各类绿地与蓝色空间的类型及其可及性。这些未发现显著关联的另一合理解释可能是：我们缺乏可用于衡量感知绿地或其质量的测量工具。近期证据表明，街道层面可见的绿化景观对心理健康的影响，可能比从航空视角测量的绿化景观更为重要（Helbich 等人，2019）。与此相关，区域层面绿地与健康结局之间缺乏关联的另一种可能原因是：绿地评估方式中存在更大的测量误差。尽管我们可以相当肯定地认为，到海岸线的欧氏距离测量是准确的，但绿地面积的百分比则依赖于若干其他假设，这些假设涉及“绿地”的定义、土地利用图的编制方式及时间（Helbich，2019）。例如，街道树木的存在已被证明与身体健康（de Vries et al.，2013）及心理健康（Taylor et al.，2015）的指标相关，但在本研究中却无法纳入考量。因此，本研究可能低估了绿地暴露效应。此外，在绿地使用方式方面，诸如设施配备、连通性、开阔步道及恢复性感知等质量因素，可能比绿地密度更具相关性（Dzhambov et al.，2018；Francis et al.，2012）。绿地与身体活动（PA）之间的关联，在不同绿地指标及缓冲区之间也可能存在差异（Klompmaker et al.，2018）。为了更好地捕捉日常生活中所访问的环境以用于体力活动研究，可选用的方法包括动态暴露评估（Helbich，2018）以及评估居住周边较大区域或缓冲区（Browning 与 Lee，2017）。然而，在本研究中，使用更大的周边区域可能仍会得出相同的结论。这一点已由Mytton 等人（2012）指出，他们使用了早期版本的 HSE 研究，其中绿色空间是在地理范围更大的“中等超级输出区”（MSOA）尺度上测量的（与本研究采用的 LSOAs 不同），并且同样未发现绿色空间与户外体力活动之间存在关联。尽管系统综述常得出结论：绿色空间与户外体力活动之间似乎存在正向但微弱的关联，但也有大量研究并未报告此类关联（van den Bosch与 Ode Sang，2017；Klompmaker 等人，2018）。最后，近期使用英国人口数据的研究表明，养狗可能是社区绿地密度与身体活动之间关联的重要调节因素（White 等，2018）；由于 HSE 研究中与身体活动数据相同的年份内并未包含养狗信息，因此我们无法在此研究中探讨这一问题，但这可能对我们的研究结果产生了混杂效应。

4.2. 研究优势、局限性及后续研究方向

本研究对该领域做出了若干贡献。首先，我们比此前许多相关研究（Markevych 等，2017）更深入地探讨了体力活动的作用——这是将住宅区蓝色与绿色空间同健康及福祉相联系的主要途径之一。通过将体力活动划分为水上运动、户外陆地活动以及室内/其他活动，我们能够更好地理解住宅社区环境及其利用方式与整体及心理健康之间的关联（Markevych 等，2017）。其次，本研究采用了包含整体及心理健康标准化可靠测量指标的大型流行病学数据集。该调查数据使得我们能够在不牺牲统计效能的前提下，对既往研究中涉及的众多潜在混杂因素进行分析校正，且本研究关于这些变量的结果与其他研究结果具有可比性（van den Berg等，2015；Gascon等，2017；Markevych等，2017）。第三，采用路径建模作为分析工具，我们在单一模型中评估了居住区蓝色与绿色空间、不同类型环境下的身体活动（PA），以及总体与心理健康之间的相互关系（如Markhevych等，2017所建议）。相较于仅分别分析这些变量的简化模型，这种方法能更好地控制中介变量、结局变量与解释变量之间的潜在混杂效应。然而，我们并未评估某些协变量（如社会经济因素 Wheeler 等，2012）是否调节沿海邻近性与身体活动或结局变量之间的关联；这些内容虽超出本研究范围，但值得未来进一步探究。第四，尽管户外身体活动——尤其是步行——在靠近蓝色空间区域的发生频率似乎略高于远离蓝色空间区域（例如Humpel等，2004），但据我们所知，此前尚无流行病学研究评估这种关联是否足以与更好的总体或心理健康感知相关联（Gascon等，2017）。

然而，本研究并非没有局限性，其中许多局限性在基于横断面调查数据的研究中十分常见。本研究无法推断因果关系，而相关证据通常较为有限（Gascon 等，2015）。根据我们的分析，可以得出以下结论：居住在靠近海岸地区的人比居住在距海岸50公里以上地区的人进行更多的户外体力活动（尤其是步行），并且感觉更健康。然而，我们无法断定是更活跃、更健康的人群更倾向于迁居至海岸附近，还是海岸邻近性促使人们更多地参与体力活动，进而促进健康。此外，在当前分析中，我们依赖于自我报告的体力活动数据，尽管有证据表明此类报告可能被夸大（Hagstromer等，2010）。尽管2008年 HSE 研究的部分参与者曾在研究期间佩戴加速度计，但我们仍无法在此使用这些数据，因为这些数据与活动信息或地理空间因素无关。即使我们当时能够在此处使用这些工具，自我报告与客观测量的身体活动数据也并非完美无缺：尽管自我报告往往存在夸大倾向，但这种高估在不同人口统计学群体及不同活动类型中相对相似；而加速度计则无法捕捉某些常见活动，如骑行，以及对于本研究可能至关重要的游泳及其他水上运动（Aresu et al.，2009a）。

本研究也受到数据提供者提供的环境数据相对粗糙的限制，这些数据对于确保受访者匿名性至关重要。因此，我们无法评估任何区域或季节性模式（Scholes 与 Mindell，2013），也无法在受访者住所周边更大区域内测量绿地或淡水要素（例如通过识别邻近区域，或在设定距离内计算这些指标），而这些信息对于身体活动行为可能更为重要（Giles-Corti 等，2005）。由于仅能分析 LSOA 层级的所有环境信息，我们未能评估可调整区域单元问题，即使用其他行政边界或缓冲区是否会产生不同结果（尽管我们发现，在使用上述更大空间单元时，绿地与身体活动方面的结果与Mytton 等（2012）的研究结果相当）。同样地，我们未能获取环境指标（如距海岸线的精确距离或绿地密度）的精确数值，而仅能采用宽泛的分类标准，这限制了更精细的分析。

最后，关于研究局限性，尽管我们掌握了所进行体力活动类型的精确信息，但我们仍无法将这些活动与特定环境或环境类型直接关联起来——这一研究重点已被Markevych 等人（2017）提出。因此，我们的体力活动分类仅基于活动本身，且范围相当广泛。某些被归类为“ζ室内/其他体力活动”的活动实际上可能在户外进行。在敏感性分析（附录表F.1）中，我们评估了环境指标与“ζ室内/其他体力活动”之间的关联，结果显示所有关联均不显著，这或许印证了我们的假设：主要的户外体力活动类型已通过“ζ陆地户外体力活动”指标（Mytton 等人，2012）充分涵盖。关于水上运动，最大的局限在于我们无法可靠地将游泳纳入该指标，因为缺乏关于游泳是在室内还是户外进行的信息。这或许部分解释了为何水上运动作为一类活动相对少见（尽管该结果与其他调查结果一致；Elliott等人，2018）；其解释方差较低，且显示出较少的

and it showed few associations with covariates and outcomes. Future studies could reduce this problem by endeavouring to ask separately about indoor and outdoor swimming, and preferably differentiate between built/artificial swimming pools and beaches.

Longitudinal studies are needed to examine the effects of moving closer to, or further from the coast, on PA levels (see White et al., 2013a for a similar approach exploring general and mental health over time). Also country-specific and contextual patterns would be useful to examine to see whether our results are replicated or contrasted in other countries and regions. Differences in cultural patterns of outdoor recreation and environmental qualities and facilities may impact how coastal, inland, and green settings are used and how their usage in turn affects general and mental health. In England, coastal settings are accessible and managed so that they are suitable for outdoor recreation and walking (Natural England, 2013). Whether similar infrastructure in other countries or regions could promote PA near coastal surroundings is worthy of investigation. Given the apparent importance of walking in coastal areas, proximity to accessible freshwater 'edges' (which may similarly represent PA opportunities such as riverside paths) may be a more appropriate inland blue space measure for future research than the area coverage that was available for this study. Finally, many epidemiological studies such as the present one do not allow for refined analysis on different aspects of environmental quality. More research efforts using mapping tools that locate PA to specific settings are needed to complement the population-level studies with more detailed analyses on the types of settings that promote better general and mental health.

4.3. Conclusion

This is the first UK study to provide clear evidence that benefits to health and wellbeing associated with living near the coast are partially mediated through outdoor physical activity. Sensitivity analyses found that walking, by far the most common type of PA and less susceptible to socio-economic or demographic differences, is more common nearer the coast, and partly accounts for why living close to the coast is related to better general and mental health. However, the different types of outdoor PA explain only a part of the coastal proximity-health relationship, indicating that other mechanisms also play a role. Having freshwater blue spaces, such as rivers and lakes near one's home was also associated with better mental health but this association was not mediated by PA, suggesting other factors could be more important. In sum, extensive analysis of two waves of the Health Survey for England, suggested that, all else equal, living near water was associated with better self-reported mental and physical health and that greater amounts of physical activity appears to account for at least part of these relationships, especially among those living near the coast.

Declaration of Competing Interest

None.

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Appendix A. Supplementary data

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显著性。与协变量及结局变量相关的研究。未来研究可通过分别探究室内与室外游泳活动，并尽可能区分人工建造/人造游泳池与海滩，以缓解这一问题。

需开展纵向研究，以探究靠近或远离海岸地区对身体活动水平的影响（关于长期观察下一般健康与心理健康的类似研究方法，可参阅White等人，2013a）。此外，针对不同国家及特定情境下的模式分析也将有助于验证本研究结果是否在其他国家及地区得以复现或出现差异。户外休闲文化模式以及环境品质与设施的差异，可能影响沿海、内陆及绿色区域的利用方式，进而影响这些区域对一般健康与心理健康的作用。在英格兰，沿海区域具备便捷的可达性与完善的管理措施，因而适宜开展户外休闲与步行活动（Natural England，2013）。其他国家或地区是否具备类似基础设施以促进沿海周边区域的身体活动，值得进一步探究。鉴于步行在沿海地区的显著重要性，对于未来研究而言，与易于利用的淡水滨缘区（此类区域亦可提供类似体力活动机会，如河岸步道）的邻近性，可能比本研究所采用的面积覆盖指标更适合作为内陆蓝色空间的衡量指标。最后，包括本研究在内的众多流行病学研究均无法对环境质量的不同维度进行精细化分析。亟需利用可将体力活动定位至特定环境场景的制图工具开展更多研究，以补充人群层面的研究，从而对促进整体及心理健康的优质环境类型进行更详尽的分析。

4.3. 结论

这是英国首项提供明确证据的研究，证实与沿海居住相关的健康及福祉获益，部分是通过户外体力活动介导实现的。敏感性分析表明：步行作为目前最普遍的体力活动类型，且较少受社会经济或人口统计学差异影响，在沿海地区更为常见；这也部分解释了为何沿海居住与整体及心理健康状况改善相关。然而，不同类型的户外体力活动仅能部分解释沿海居住与健康之间的关联，提示其他机制亦发挥作用。拥有家庭附近的淡水蓝色空间（如河流与湖泊）同样与心理健康状况改善相关，但该关联并非由体力活动介导，表明其他因素可能更为关键。综上所述，通过对《英格兰健康调查》两轮数据的全面分析发现：在其他条件相同的情况下，近水居住与自我报告的身心健康状况改善相关；且更多体力活动量似乎至少可部分解释这些关联，尤其在沿海居民中更为显著。

利益冲突声明

没有一个.

致谢

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附录 A：补充数据

本文的补充数据可在以下网址在线获取：https://doi.org/10.1016/j.envint.2019.105016。

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